Guanhao Wang

gwang284@wisc.edu | +1 6084403924 | +86 19921697926

Portfolio | GitHub

Computer science undergraduate focused on game development, environmental storytelling, and physical interaction.

EDUCATION

University of Wisconsin–Madison June 2025 to present
Computer Science Undergraduate · Expected graduation May 2027 · GPA 3.724/4.000

The Pennsylvania State University August 2023 to June 2025
Undergraduate Study in Computer Science · GPA 3.94/4.00 · Transferred to UW–Madison

GAMES AND INTERACTIVE PROJECTS

Coal Echoes 2026
Narrative Exploration Game · In development · Unity 6, HDRP, C#, Blender

- Designed a first person exploration game about family memories of an abandoned coal plant; built photography, album, scrapbook, and object inspection systems, enabling object discovery through the camera.
- Built industrial environments with DEM terrain, a moving cableway, sandstorms, and heat haze; generated main menu newspaper photo materials at runtime from live HDRP Custom Pass camera captures, using HLSL halftone logic adapted from Daniel Ilett (MIT).

Project Fireflies 2026
Interactive Light Installation · Completed · ESP32, Arduino, Unity, UDP

- Programmed 420 LEDs across three swarms, using radar and light sensors to translate movement and ambient light into firefly disturbance and synchronization.
- Implemented two independently responsive zones, one ambient swarm, and a UDP interface for Unity synchronization commands.

Seafloor Revival July to August 2025
GMTK Game Jam 2025 · UI Development and Art Integration · Unity, C#

- Built all game UI and related C# behavior, integrated interface art in a six person team, and contributed to a released browser prototype of a 2.5D underwater building game.

Mooncake February to April 2025
Interactive Graphics · Three.js, JavaScript

- Developed a Three.js particle system and web demo with live controls for particle count and size; documented the implementation in a technical report.

PROFESSIONAL EXPERIENCE

IQVIA July to August 2024
Technology Solutions Intern · Shanghai, China

- Wrote and maintained 100+ pages of user guides for a pharmaceutical WeChat Mini Program upgrade, covering two user types, four scenarios, and three device contexts.
- Performed user acceptance testing and reported 10+ defects in Jira with reproduction steps; organized 1,000+ data tags into a three level hierarchy for user filtering.

PUBLICATION

Human Computer Interaction in Indie Games 2023
Sole Author

- Analyzed implicit guidance, environmental storytelling, and player attachment in FAR Lone Sails; published in Lecture Notes in Education Psychology and Public Media, vol. 18, pp. 194 to 199.

SKILLS AND LANGUAGES

Games and interaction Unity, C#, HDRP, uGUI, Shader Graph, Blender, ESP32, Arduino

Programming and tools Python, Java, C, JavaScript, Three.js, Git, Jira

Languages Mandarin native, English fluent, Japanese basic